

LAMPIRAN

Lampiran 1

Subkultur Isolat Bakteri

Pembuatan Media *Nutrient Agar*

Menyiapkan 10 cawan petri dibungkus dengan kertas dan erlenmeyer 250 ml

<i>Agar bacteriological</i> Lp 0011	1,5 g
<i>Nutrient broth agar</i>	1,3 g
Aquadest	100 ml

Homogenkan, erlenmeyer ditutup dengan aluminium foil, di-*autoclave* selama 20 menit dengan suhu 121°C. Kemudian *microwave* dengan kekuatan *high* (150°-200°C) selama 3-5 menit.

Lampiran 2

Polymerase Chain Reaction (PCR) 16S rRNA

Pembuatan produk PCR :

Primer 27F (5'-AGAGTTTGATCMTGGCTCAG-3')	1 µl
Primer 1492R (5'-TACGGYTACCTTGTTACGACTT-3')	1 µl
DNA <i>template</i>	3 µl
ddH ₂ O	7,5 µl
<i>Mastermix bioline</i>	12,5 µl
Jumlah	25 µl

Kemudian *thermocycle* diatur dengan

<i>Pra-denaturation</i>	96°C	1 menit	
<i>Denaturation</i>	96°C	15 detik	(Siklus
<i>Annealing</i>	55°C	15 detik	sebanyak
<i>Elongation</i>	72°C	10 detik	34 kali)
<i>Extention</i>	72°C	12 menit	
<i>Cooling down</i>	4°C	Tanpa batas	

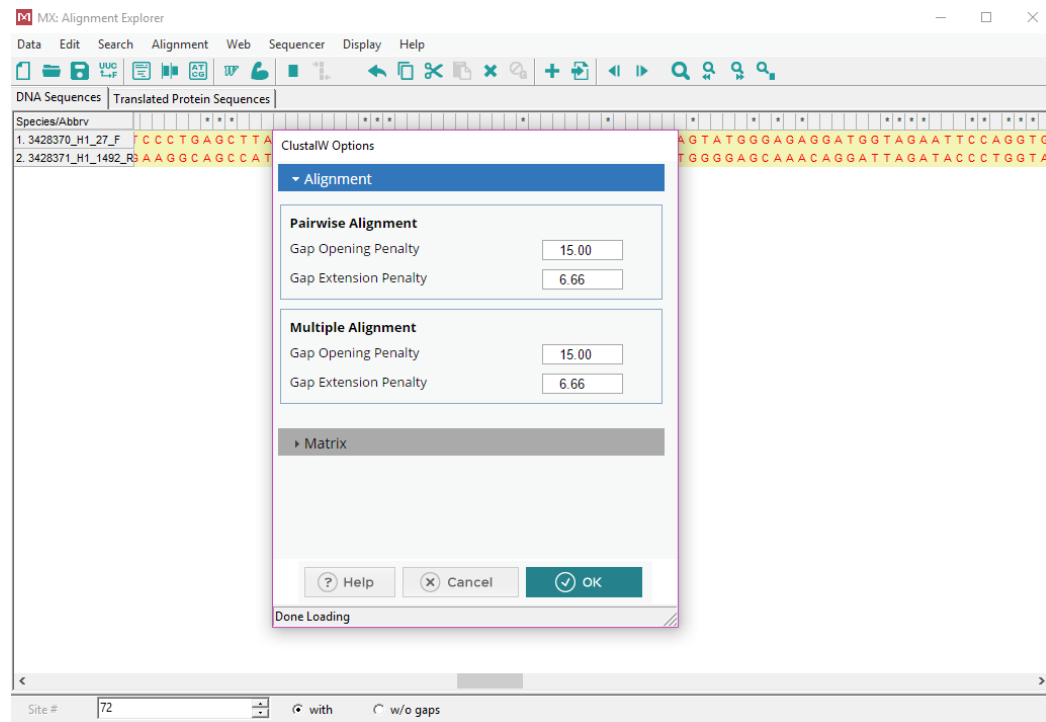
Lampiran 3

Sekuensing DNA

Grafik dari hasil sekuensing primer *forward* dan *reverse*



Proses editing sekuens *forward* dan *reverse* dengan pensejajaran menggunakan program alignment ClustalW



Lampiran 4

Program Basic Local Alignment Search Tools (BLAST)

Memasukkan data awal

MEGA Web Browser: Nucleotide BLAST: Search nucleotide databases using a nucleotide query

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi?CMD=Web&LAYOUT=OneWindows&AUTC

Nucleotide BLAST: Search nucleotide databases using a nucleotide query

NIH U.S. National Library of Medicine NCBI National Center for Biotechnology Information Sign in to NCBI

BLAST » blastn suite Home Recent Results Saved Strategies Help

Standard Nucleotide BLAST

blastn blastp blastx tblastn tblastx

BLASTN programs search nucleotide databases using a nucleotide query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s)

Clear Query subrange

From To

Or, upload file Choose File No file chosen

Job Title

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database

Human genomic + transcript Mouse genomic + transcript Others (nr etc.)

16S ribosomal RNA sequences (Bacteria and Archaea)

Organism

Optional

Enter organism name or id-completions will be suggested

Exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown

Sequences from type material

Limit to

Optional

Entrez Query

Optional

Enter an Entrez query to limit search

You Like Create custom database

We are beta testing a New Results page

Click here if you would like to see your results in the new format. You can always switch back to the Traditional Results page.

Hasil dari BLAST

H1

MEGA Web Browser: NCBI Blast:Nucleotide Sequence

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NCBI Blast:Nucleotide Sequence | Acinetobacter schindleri 16S rRNA gene, strain LUH 4591 - Nucleotide - NCBI

Select all sequences selected

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☒	Acinetobacter schindleri strain SGAir0122 chromosome, complete genome	2427	16880	100%	0.0	98.01%	gi1530819886 CP025618.2
☐	Uncultured bacterium gene for 16S ribosomal RNA, partial sequence, clone PW72	2427	2427	100%	0.0	98.01%	gi456359297 AB745643.1
☐	Acinetobacter sp. JNG7 16S ribosomal RNA gene, partial sequence	2425	2425	100%	0.0	98.01%	gi443609748 KC503919.1
☒	Acinetobacter schindleri strain SNSK 752 16S ribosomal RNA gene, partial sequence	2422	2422	100%	0.0	97.94%	gi1280931553 MG584984.1
☐	Acinetobacter johnsonii strain 3-1 16S ribosomal RNA gene, partial sequence	2422	2422	100%	0.0	97.94%	gi183230088 EU594557.1
☐	Acinetobacter sp. strain TAGEM15-70-B21 16S ribosomal RNA gene, partial sequence	2420	2420	100%	0.0	97.94%	gi11314869675 MG685814.1
☐	Acinetobacter schindleri strain YNB103 16S ribosomal RNA gene, partial sequence	2420	2420	100%	0.0	97.94%	gi374431088 JQ039984.1
☐	Acinetobacter sp. 4-2 16S ribosomal RNA gene, partial sequence	2420	2420	100%	0.0	97.94%	gi183230089 EU594561.1
☐	Acinetobacter schindleri strain ZSR6 16S ribosomal RNA gene, partial sequence	2418	2418	99%	0.0	98.00%	gi1115585659 KY305021.1
☐	Acinetobacter calcoaceticus strain WH-B2 16S ribosomal RNA gene, partial sequence	2416	2416	100%	0.0	97.87%	gi1182953891 KY963634.1
☐	Acinetobacter schindleri strain SJC43 16S ribosomal RNA gene, partial sequence	2416	2416	100%	0.0	97.87%	gi1033207264 KC301302.1
☐	Acinetobacter schindleri strain SJC6 16S ribosomal RNA gene, partial sequence	2416	2416	100%	0.0	97.87%	gi1033207263 KC301301.1
☐	Acinetobacter sp. CIP 101934 16S ribosomal RNA gene, partial sequence	2416	2416	100%	0.0	97.87%	gi375161309 JQ638573.1
☐	Acinetobacter sp. EH 28 16S ribosomal RNA gene, partial sequence	2416	2416	100%	0.0	97.87%	gi189031500 EU703817.1
☐	Acinetobacter johnsonii strain TAGEM15-70-B20 16S ribosomal RNA gene, partial sequence	2414	2414	99%	0.0	98.00%	gi11314818063 MG685688.1
☐	Acinetobacter sp. BZC6 16S ribosomal RNA gene, partial sequence	2414	2414	100%	0.0	97.86%	gi1043446191 KU984709.1
☐	Acinetobacter schindleri strain RM-13 16S ribosomal RNA gene, partial sequence	2412	2412	99%	0.0	98.00%	gi925687563 KR029982.1
☐	Acinetobacter schindleri strain XJ-10 16S ribosomal RNA gene, partial sequence	2410	2410	100%	0.0	97.79%	gi1279339940 MG551527.1
☐	Acinetobacter schindleri strain ACE, complete genome	2410	16876	100%	0.0	97.80%	gi1134946078 CP015615.1
☐	Acinetobacter schindleri strain MCD403 16S ribosomal RNA gene, partial sequence	2410	2410	100%	0.0	97.80%	gi1125677803 KY303037.1
☐	Uncultured bacterium clone 624forB_31 16S ribosomal RNA gene, partial sequence	2410	2410	100%	0.0	97.80%	gi447603157

Feedback

H3

MEGA Web Browser: NCBI Blast:dbj|C482254.1|

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NCBI Blast:dbj|C482254.1|

select all 0 sequences selected

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☐	Stenotrophomonas sp. strain R50 16S ribosomal RNA gene, partial sequence	2603	2603	100%	0.0	99.86%	gi1452158939 MH773411.1
☐	Stenotrophomonas sp. enrichment culture clone CW-4Y 16S ribosomal RNA gene, partial sequence	2603	2603	100%	0.0	99.86%	gi356578596 JN995250.1
☐	Stenotrophomonas sp. M2T2B9 16S ribosomal RNA gene, partial sequence	2599	2599	100%	0.0	99.79%	gi261291229 GQ246699.1
☐	Stenotrophomonas sp. M2 16S ribosomal RNA gene, partial sequence	2599	2599	100%	0.0	99.79%	gi209972642 FJ357297.1
☐	Stenotrophomonas sp. LW-34 16S ribosomal RNA gene, partial sequence	2597	2597	100%	0.0	99.79%	gi928241936 KR258782.1
☐	Stenotrophomonas sp. LW-30 16S ribosomal RNA gene, partial sequence	2597	2597	100%	0.0	99.79%	gi928241932 KR258778.1
☐	Stenotrophomonas sp. enrichment culture clone 2 16S ribosomal RNA gene, partial sequence	2597	2597	100%	0.0	99.79%	gi610390753 KJ191559.1
☐	Stenotrophomonas sp. strain Gamma-29 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi1439156953 MH703460.1
☐	Stenotrophomonas sp. N054 16S ribosomal RNA gene, partial sequence	2593	2593	99%	0.0	99.86%	gi460479466 KC252737.1
☐	Stenotrophomonas maltophilia strain YSP48 16S ribosomal RNA gene, partial sequence	2593	2593	99%	0.0	99.79%	gi335893087 JF894170.1
☐	Stenotrophomonas sp. SP3 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi239735952 FJ976656.1
☐	Uncultured bacterium clone ctg1_TOPO1-17 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi189182428 EJ1708503.1
☐	Uncultured bacterium clone SZ3_289 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi148729971 EF572190.1
☐	Stenotrophomonas sp. YC-1 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi106636140 DQ537219.1
☐	Stenotrophomonas sp. Dsp-4 16S ribosomal RNA gene, partial sequence	2593	2593	100%	0.0	99.72%	gi94470220 DQ482654.1
☐	Stenotrophomonas sp. LMG 19633 partial 16S rRNA gene, strain LMG 19633	2593	2593	100%	0.0	99.72%	gi18250908 AJ300772.1
☐	Stenotrophomonas maltophilia strain AsPCI 2.3 16S ribosomal RNA gene, partial sequence	2591	2591	100%	0.0	99.72%	gi1569655785 MK503992.1
☐	Stenotrophomonas sp. strain 16H1E7 16S ribosomal RNA gene, partial sequence	2591	2591	99%	0.0	99.72%	gi1240338367 MF872594.1
☐	Stenotrophomonas sp. SCU-B2602 16S ribosomal RNA gene, partial sequence	2590	2590	99%	0.0	99.79%	gi595250786 KJ000871.1
☐	Stenotrophomonas sp. BO 16S ribosomal RNA gene, partial sequence	2590	2590	100%	0.0	99.65%	gi7021315 AF
☐	Bacterium 14A9 16S ribosomal RNA gene, partial sequence	2590	2590	100%	0.0	99.65%	gi83367131 D

Feedback

H5

MEGA Web Browser: NCBI Blast(3) - dbj|C482253.1|

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NCBI Blast(3) - dbj|C482253.1| Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene - Nucleotide - NCBI

select all 0 sequences selected

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☐	Stenotrophomonas sp. strain R50 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi1452158939 MH773411.1
☐	Stenotrophomonas sp. strain Gamma-29 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi1439156953 MH703460.1
☐	Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi1261492573 MG205630.1
☐	Stenotrophomonas sp. strain JU63 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi1041940535 KU375545.1
☐	Stenotrophomonas sp. LW-34 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi928241936 KR258782.1
☐	Stenotrophomonas sp. LW-30 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi928241932 KR258778.1
☐	Stenotrophomonas sp. enrichment culture clone 2 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi610390753 KJ191559.1
☐	Bacterium NC1-3 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi523506975 KC905047.1
☐	Bacterium NC1-2 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi523506969 KC905046.1
☐	Stenotrophomonas sp. R063 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479622 KC252893.1
☐	Stenotrophomonas sp. N064 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479466 KC252737.1
☐	Stenotrophomonas sp. P042aR 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479509 KC252780.1
☐	Stenotrophomonas sp. P026NS 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479493 KC252764.1
☐	Stenotrophomonas sp. P002 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479463 KC252740.1
☐	Stenotrophomonas sp. N060 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479462 KC252733.1
☐	Stenotrophomonas sp. N059a 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479460 KC252731.1
☐	Stenotrophomonas sp. N054 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479455 KC252726.1
☐	Stenotrophomonas sp. N025 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479427 KC252698.1
☐	Stenotrophomonas sp. N006NS 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479408 KC252679.1
☐	Stenotrophomonas sp. N006NR 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi460479407
☐	Stenotrophomonas sp. enrichment culture clone CW-4Y 16S ribosomal RNA gene, partial sequence	2078	2078	87%	0.0	97.69%	gi356578596

Feedback

T3

MEGA Web Browser: NCBI Blast:dbj|C482252.1|

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NCBI Blast:dbj|C482252.1|

Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☐ Uncultured bacterium clone NY7_9_16S ribosomal RNA gene, partial sequence	2483	2483	100%	0.0	98.92%	gi1003096816 KT851827.1
☐ Uncultured bacterium clone Ark7_20_16S ribosomal RNA gene, partial sequence	2483	2483	100%	0.0	98.92%	gi1003096791 KT851802.1
☐ Uncultured bacterium clone NN50_16S ribosomal RNA gene, partial sequence	2483	2483	100%	0.0	98.92%	gi359551974 JN869095.1
☐ Uncultured bacterium clone Dianchi-11_16S ribosomal RNA gene, partial sequence	2483	2483	100%	0.0	98.92%	gi319998925 HQ324859.1
☐ Uncultured bacterium clone OP27_16S ribosomal RNA gene, partial sequence	2479	2479	100%	0.0	98.85%	gi960516078 KT283068.1
☐ Uncultured bacterium clone S66_16S ribosomal RNA gene, partial sequence	2479	2479	100%	0.0	98.85%	gi359551579 JN868745.1
☐ Rheinheimera aquatica strain BK-127_16S ribosomal RNA gene, partial sequence	2477	2477	100%	0.0	98.85%	gi1024638470 KU360716.1
☐ Gamma proteobacterium BIWA55 gene for 16S ribosomal RNA, partial sequence	2477	2477	100%	0.0	98.85%	gi1149120662 C217442.1
☐ Gamma proteobacterium RIWA54 gene for 16S ribosomal RNA, partial sequence	2477	2477	100%	0.0	98.85%	gi1149120661 C217441.1
☐ Uncultured bacterium clone Ark9_10_16S ribosomal RNA gene, partial sequence	2477	2477	100%	0.0	98.85%	gi1003096796 KT851807.1
☐ Rheinheimera sp. TAPG2 partial 16S rRNA gene, strain TAPG2 isolate stream	2477	2477	100%	0.0	98.85%	gi1081426098 LT627667.1
☐ Uncultured bacterium clone T0-An-20C-18_16S ribosomal RNA gene, complete sequence	2477	2477	100%	0.0	98.85%	gi399107268 JX105532.1
☐ Uncultured bacterium clone DP3_2_2_16S ribosomal RNA gene, partial sequence	2477	2477	100%	0.0	98.85%	gi255739771 FJ612134.1
☐ Uncultured bacterium clone MC1_16S_55_16S ribosomal RNA gene, partial sequence	2477	2477	100%	0.0	98.85%	gi194478148 EU662624.1
☐ Uncultured bacterium clone MC1_16S_54_16S ribosomal RNA gene, partial sequence	2477	2477	100%	0.0	98.85%	gi194478147 EU662623.1
☐ Uncultured bacterium clone NY7_6_16S ribosomal RNA gene, partial sequence	2471	2471	100%	0.0	98.78%	gi1003096817 KT851828.1
☐ Uncultured bacterium clone Ark7_16_16S ribosomal RNA gene, partial sequence	2471	2471	100%	0.0	98.78%	gi1003096790 KT851801.1
☐ Uncultured bacterium clone Ark7_15_16S ribosomal RNA gene, partial sequence	2471	2471	100%	0.0	98.78%	gi1003096789 KT851800.1
☐ Uncultured bacterium clone Ark7_18_16S ribosomal RNA gene, partial sequence	2471	2471	100%	0.0	98.78%	gi1003096788 KT851799.1
☐ Rheinheimera sp. TH212_16S ribosomal RNA gene, partial sequence	2468	2468	99%	0.0	98.98%	gi1189986184 KT826392.1
☐ Uncultured bacterium clone EHB-P0732 #02_16S ribosomal RNA gene, partial sequence	2468	2468	100%	0.0	98.71%	gi1152261168

Feedback

Hasil pensejajaran sekuens pada GenBank

MEGA Web Browser: NCBI Blast:Nucleotide Sequence

File Edit View Navigate Help

https://blast.ncbi.nlm.nih.gov/Blast.cgi#alnHdr_219857516

NCBI Blast:Nucleotide Sequence

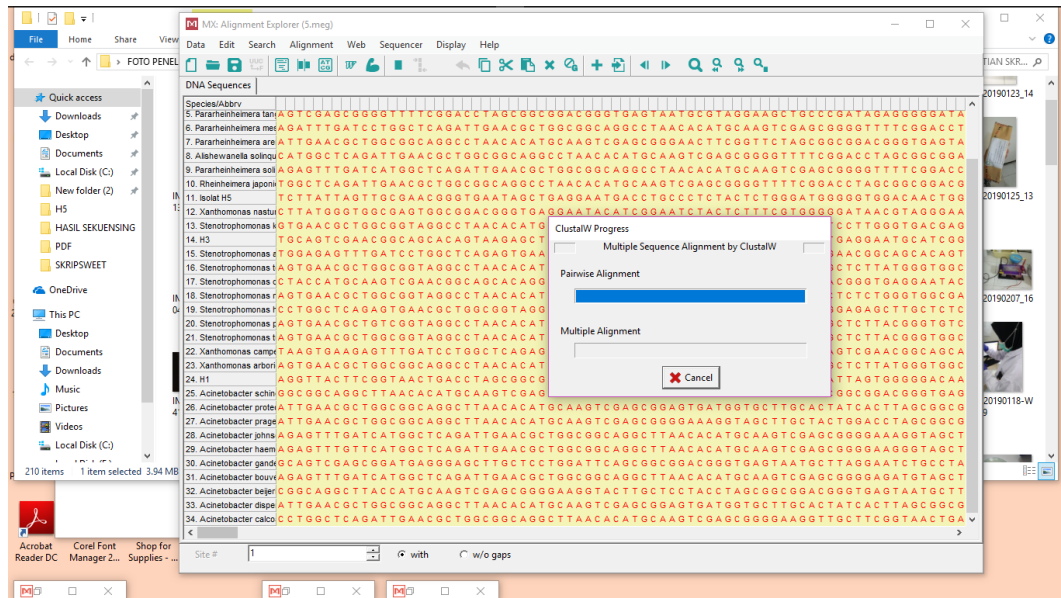
Score	Expect	Identities	Gaps	Strand
2571 bits(1392)	0.0	1410/1418(99%)	3/1418(0%)	Plus/Minus
Query 1	TGC - AGTCGAACGGCAGCACAGTAAAGACTTGCTCTTACGGGTGGCGAGTGGCGGACGGG	59		
Sbjct 52	TGCAAGTCGAACGGCAGCACAGTAAAGACTTGCTCTTACGGGTGGCGAGTGGCGGACGGG	111		
Query 60	TGAGGAATGCATCGGAATCTACTCTGTCTGTGGGGGATAACGTAGGGAAATCTACGCTAAT	119		
Sbjct 112	TGAGGAATGCATCGGAATCTACTCTGTCTGTGGGGGATAACGTAGGGAAATCTACGCTAAT	171		
Query 120	ACCGCATACGACCTACGGGTGAAGCAGGGGATCTTCGGACCTTGCGGATTTGAATGAGC	179		
Sbjct 172	ACCGCATACGACCTACGGGTGAAGCAGGGGATCTTCGGACCTTGCGGATTTGAATGAGC	231		
Query 180	CGATGCCCGATTAGCTAGTTGGCGGGGTAAAGAGCCCAACCAAGGCGACGATCGGTAGCTGG	239		
Sbjct 232	CGATGCCCGATTAGCTAGTTGGCGGGGTAAAGAGCCCAACCAAGGCGACGATCGGTAGCTGG	291		
Query 240	TCTGAGAGGATGATCAGCCACACTGGAACTGAGACGGTCCAGACTCTACGGGAGGCA	299		
Sbjct 292	TCTGAGAGGATGATCAGCCACACTGGAACTGAGACGGTCCAGACTCTACGGGAGGCA	351		
Query 300	GCAGTGGGGAATATTGGACAATGGGCGCAAGCCTGATCCAGCCATACCGGTGGGTGAAG	359		
Sbjct 352	GCAGTGGGGAATATTGGACAATGGGCGCAAGCCTGATCCAGCCATACCGGTGGGTGAAG	411		
Query 360	AAGGCTTCGGGTTGTAAGCCCTTTTGTGGGAAAGAAAGCATTTCGGTTAATACCCGA	419		
Sbjct 412	AAGGCTTCGGGTTGTAAGCCCTTTTGTGGGAAAGAAAGCAGCCGGTTAATACCCGG	471		
Query 420	TTGTTCTGACGGTACCAAGAATAAGCACCAGCTAACTTCGTGCCAGCAGCCGCGGTAA	479		
Sbjct 472	TTGTTCTGACGGTACCAAGAATAAGCACCAGCTAACTTCGTGCCAGCAGCCGCGGTAA	531		
Query 480	TACGAAGGGTGCAAGGTTACTCGGAATTACTGGGCGTAAAGCGTGCGTAGGTGGTTGTT	539		
Sbjct 532	TACGAAGGGTGCAAGGTTACTCGGAATTACTGGGCGTAAAGCGTGCGTAGGTGGTTGTT	591		
Query 540	TAAGTCTGTCGTGAAAGCCCTGGGCTCAACCTGGGAATTGCGATGGAACTGGGCGACTA	599		
Sbjct 592	TAAGTCTGTCGTGAAAGCCCTGGGCTCAACCTGGGAATTGCGATGGAACTGGGCGACTA	651		

Feedback

Lampiran 5

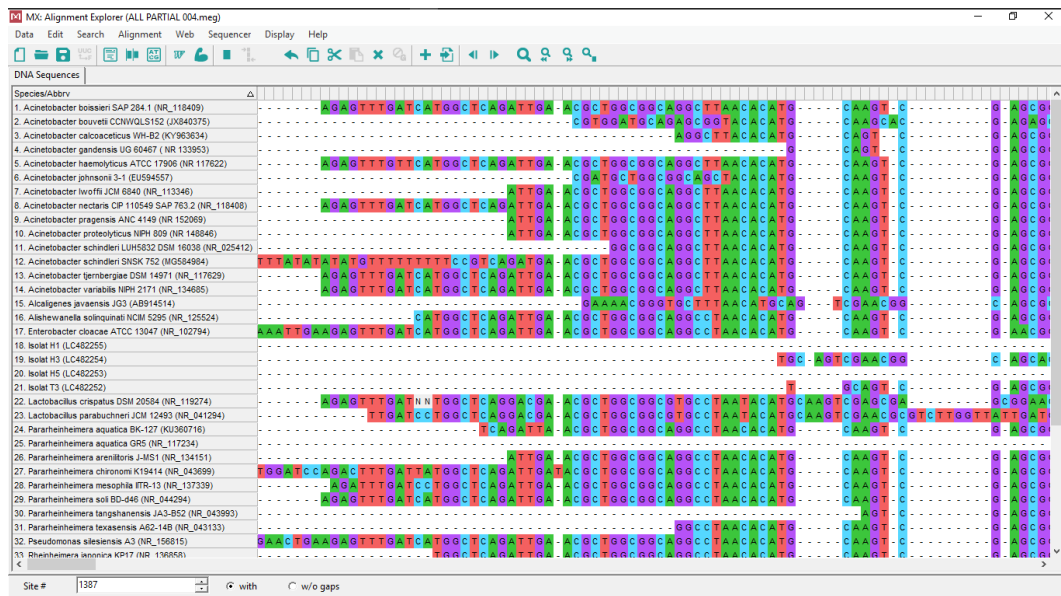
Multiple Sequences Alignment (Pensejajaran multiple atau banyak sekuens)

Program ClustalW



Hasil *Multiple Sequences Alignment*

Garis-garis merupakan *gap* atau jarak yang memisahkan



Lampian 6

Data telah didPOSIT ke *GenBank* dengan nomor akses

H1 (LC482255)

https://www.ncbi.nlm.nih.gov/nuccore/LC482255

NCBI Blast:dbj|LC482253.1| Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene - Nucleotide - NCBI Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequen - Nucleotide - NC

Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequence

GenBank: LC482255.1
[FASTA](#) [Graphics](#)

Go to:

LOCUS	LC482255	1405 bp	DNA	linear	BCT 18-MAY-2019
DEFINITION	Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequence.				
ACCESSION	LC482255				
VERSION	LC482255.1				
KEYWORDS	.				
SOURCE	Acinetobacter schindleri				
ORGANISM	Acinetobacter schindleri Bacteria; Proteobacteria; Gammaproteobacteria; Pseudomonadales; Moraxellaceae; Acinetobacter.				
REFERENCE	1				
AUTHORS	Iskandar,A.U., Sabrina,A.N., Arifiani,N., Sulistyaningtyas,A.R., Muchlissin,S.I., Darmawati,S. and Ethica,S.N.				
TITLE	Bioprospecting Indigenous Proteolytic Bacteria from Liquid Biomedical Waste of Community Health Center as Bioremediation Agent				
JOURNAL	Unpublished				
REFERENCE	2 (bases 1 to 1405)				
AUTHORS	Ethica,S.N.				
TITLE	Direct Submission				
JOURNAL	Submitted (14-MAY-2019) Contact:Stalis Norma Ethica Universitas Muhammadiyah Semarang, Medical Laboratory Technology; Jalan Kedungmundu Raya No. 18, Semarang 60273, Indonesia URL :http://unimus.ac.id				
FEATURES	Location/Qualifiers				
source	1..1405 /organism="Acinetobacter schindleri" /mol_type="genomic DNA" /strain="H1" /isolation_source="Biomedical wastewater" /db_xref="taxon:108981" /country="Indonesia" /lat_lon="6.9798 S 110.4576 E"				

Basic Features

☐ All features

☒ Gene, RNA, and CDS features

Display options

☒ Show sequence

☐ Show reverse complement

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Recent activity

Acinetobacter schindleri H1 ribosomal RNA, partial seq

Xanthomonas campestris s 33913 16S ribosomal RNA,

Lactobacillus crispatus stra 16S ribosomal RNA, partial

Lactobacillus amylolyticus (biomol_rna[PROP]) (3)

Enterobacter cloacae strair 16S ribosomal RNA, compl

H3 (LC482254)

MEGA Web Browser: Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial se - Nucleotide - NCBI

File Edit View Navigate Help

https://www.ncbi.nlm.nih.gov/nucore/LC482254 + Add To Alignment

NCBI Blast:dbj|LC482253.1| Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene - Nucleotide - NCBI Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial se - Nucleotide - NCBI

Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial sequence

GenBank: LC482254.1
[FASTA](#) [Graphics](#)

Go to:

LOCUS LC482254 1416 bp DNA linear BCT 18-MAY-2019
 DEFINITION Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial sequence.
 ACCESSION LC482254
 VERSION LC482254.1
 KEYWORDS .
 SOURCE Stenotrophomonas maltophilia
 ORGANISM [Stenotrophomonas maltophilia](#)
 Bacteria; Proteobacteria; Gammaproteobacteria; Xanthomonadales; Xanthomonadaceae; Stenotrophomonas; Stenotrophomonas maltophilia group.
 REFERENCE 1
 AUTHORS Iskandar,A.U., Sabrina,A.N., Arifiani,N., Sulistyningtyas,A.R., Muchlissin,S.I., Darmawati,S. and Ethica,S.N.
 TITLE Bioprospecting Indigenous Proteolytic Bacteria from Liquid Biomedical Waste of Community Health Center as Bioremediation Agent
 JOURNAL Unpublished
 REFERENCE 2 (bases 1 to 1416)
 AUTHORS Ethica,S.N.
 TITLE Direct Submission
 JOURNAL Submitted (14-MAY-2019) Contact:Stalis Norma Ethica Universitas Muhammadiyah Semarang, Medical Laboratory Technology; Jalan Kedungmundu Raya No. 18, Semarang 60273, Indonesia URL :http://unimus.ac.id
 FEATURES
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 Location/Qualifiers
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Basic Features
☐ All features
☒ Gene, RNA, and CDS features only
Display options
☒ Show sequence
☐ Show reverse complement
 Update View

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Recent activity

Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial sequence Nucleotide

Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequence Nucleotide

Xanthomonas campestris strain ATCC 33913 16S ribosomal RNA, complete Nucleotide

Lactobacillus crispatus strain DSM 20584 16S ribosomal RNA, partial sequence Nucleotide

Lactobacillus amylolyticus 16S rRNA AND (biomol_rna[PROP]) (3) Nucleotide

H5 (LC482253)

MEGA Web Browser: Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial - Nucleotide - NCBI

File Edit View Navigate Help

https://www.ncbi.nlm.nih.gov/nucleotide/LC482253

NCBI Blast:dbj|LC482253.1| Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene - Nucleotide - NCBI Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial - Nucleotide - NCBI

Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial sequence

GenBank: LC482253.1

[FASTA](#) [Graphics](#)

[Go to:](#) ☐

LOCUS LC482253 1393 bp DNA linear BCT 18-MAY-2019

DEFINITION Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial sequence.

ACCESSION LC482253

VERSION LC482253.1

KEYWORDS .

SOURCE Stenotrophomonas acidaminiphila

ORGANISM [Stenotrophomonas acidaminiphila](#)
Bacteria; Proteobacteria; Gammaproteobacteria; Xanthomonadales; Xanthomonadaceae; Stenotrophomonas.

REFERENCE 1

AUTHORS Iskandar,A.U., Sabrina,A.N., Arifiani,N., Sulistyaningtyas,A.R., Muchlissin,S.I., Darmawati,S. and Ethica,S.N.

TITLE Bioprospecting Indigenous Proteolytic Bacteria from Liquid Biomedical Waste of Community Health Center as Bioremediation Agent

JOURNAL Unpublished

REFERENCE 2 (bases 1 to 1393)

AUTHORS Ethica,S.N.

TITLE Direct Submission

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FEATURES

source Location/Qualifiers

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Basic Features

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Display options

☒ Show sequence

☐ Show reverse complement

[Update View](#)

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Recent activity

[Turn Off](#) [Clear](#)

- Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial sequ Nucleotide
- Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial sequen Nucleotide
- Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequence Nucleotide
- Xanthomonas campestris strain ATCC 33913 16S ribosomal RNA, complet Nucleotide
- Lactobacillus crispatus strain DSM 20584 16S ribosomal RNA, partial sequen Nucleotide

T3 (LC482252)

MEGA Web Browser: Pararheinheimera aquatica T3 gene for 16S ribosomal RNA, partial seque - Nucleotide - NCBI

File Edit View Navigate Help

https://www.ncbi.nlm.nih.gov/nuccore/LC482252

NCBI Blast:dbj|LC482253.1| Stenotrophomonas acidaminiphila strain JR3/1-12 16S ribosomal RNA gene - Nucleotide - NCBI Pararheinheimera aquatica T3 gene for 16S ribosomal RNA, partial seque - Nucleotide - NCBI

Pararheinheimera aquatica T3 gene for 16S ribosomal RNA, partial sequence

GenBank: LC482252.1

[FASTA](#) [Graphics](#)

Go to: ☐

LOCUS LC482252 1389 bp DNA linear BCT 18-MAY-2019

DEFINITION Pararheinheimera aquatica T3 gene for 16S ribosomal RNA, partial sequence.

ACCESSION LC482252

VERSION LC482252.1

KEYWORDS .

SOURCE Pararheinheimera aquatica

ORGANISM [Pararheinheimera aquatica](#)
Bacteria; Proteobacteria; Gammaproteobacteria; Chromatiales; Chromatiaceae; Pararheinheimera.

REFERENCE 1

AUTHORS Iskandar,A.U., Sabrina,A.N., Arifiani,N., Sulistyaningtyas,A.R., Muchlissin,S.I., Darmawati,S. and Ethica,S.N.

TITLE Bioprospecting Indigenous Proteolytic Bacteria from Liquid Biomedical Waste of Community Health Center as Bioremediation Agent

JOURNAL Unpublished

REFERENCE 2 (bases 1 to 1389)

AUTHORS Ethica,S.N.

TITLE Direct Submission

JOURNAL Submitted (14-MAY-2019) Contact:Stalis Norma Ethica Universitas Muhammadiyah Semarang, Medical Laboratory Technology; Jalan Kedungmundu Raya No. 18, Semarang 60273, Indonesia URL :http://unimus.ac.id

FEATURES

source Location/Qualifiers

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Customize view

Basic Features

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☒ Gene, RNA, and CDS features only

Display options

☒ Show sequence

☐ Show reverse complement

Update View

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Recent activity

[Turn Off](#) [Clear](#)

Pararheinheimera aquatica T3 gene for 16S ribosomal RNA, partial sequence Nucleotide

Stenotrophomonas acidaminiphila H5 gene for 16S ribosomal RNA, partial sequence Nucleotide

Stenotrophomonas maltophilia H3 gene for 16S ribosomal RNA, partial sequence Nucleotide

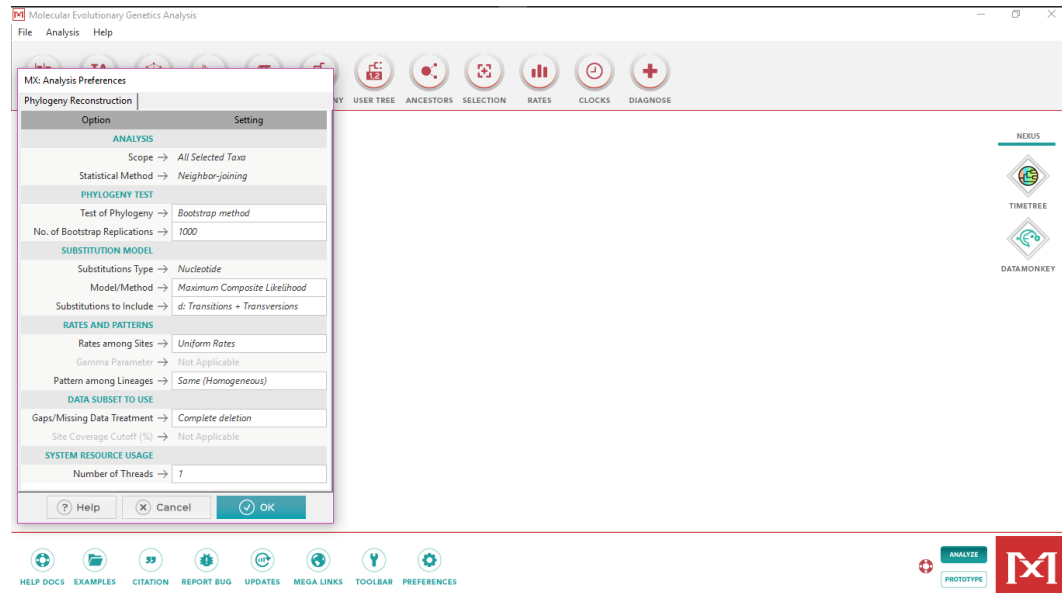
Acinetobacter schindleri H1 gene for 16S ribosomal RNA, partial sequence Nucleotide

Xanthomonas campestris strain ATCC 33913 16S ribosomal RNA, complete Nucleotide

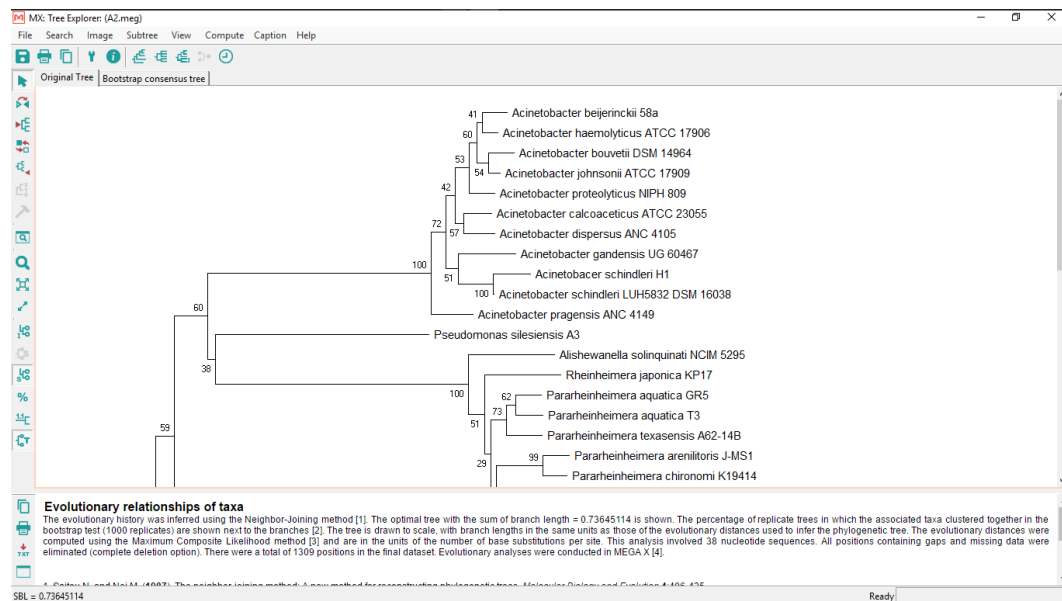
Lampiran 7

Konstruksi Pohon Filogenetik

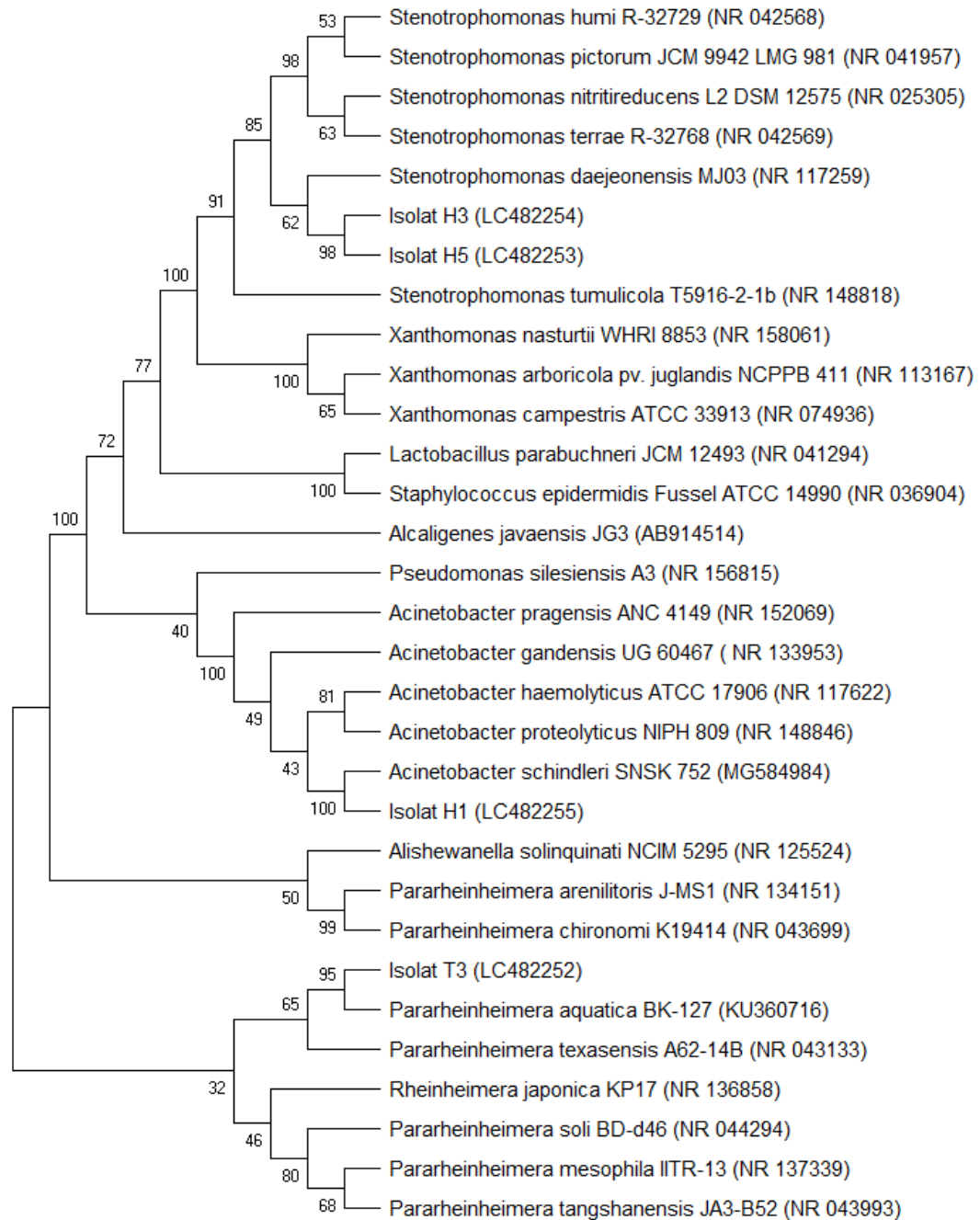
Metode Neighbor-Joining Analisis Bootstrap dengan pengulangan 1000 kali

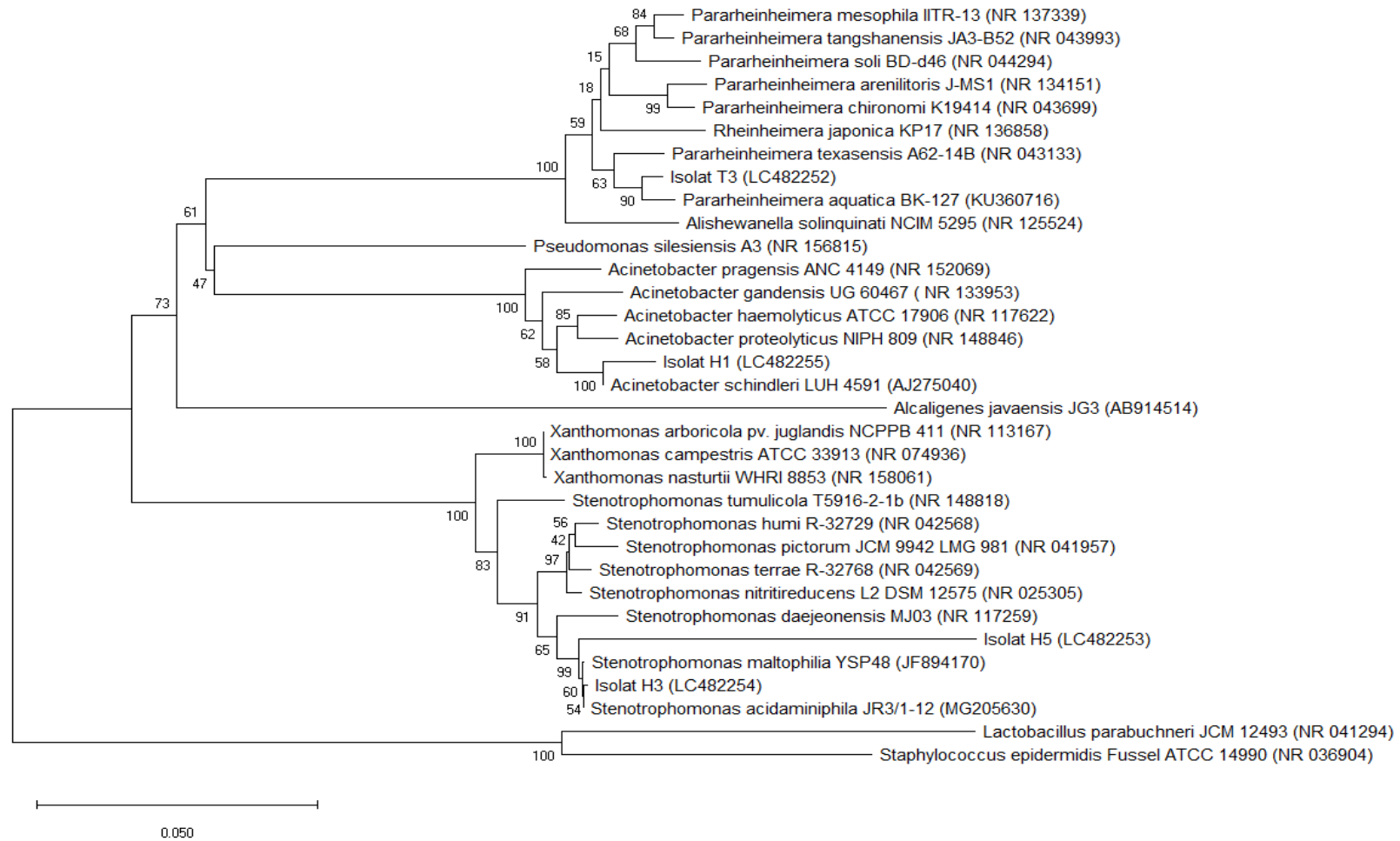


Hasil gambar pohon filogenetik



Gambar Pohon filogenetika





Gambar original pohon filogenetik 4 Isolat bakteri

Lampiran 8

Dokumentasi selama penelitian di Laboratorium Oseanografi Fakultas Kelautan dan Ilmu Perikanan (FKIP) Universitas Diponegoro



Hasil Isolasi DNA



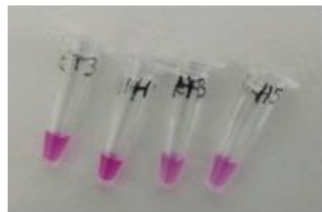
Kemurnian Isolasi DNA dengan
NanoDrop 2000



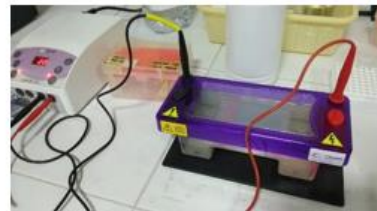
Primer Mastermix dan ddH₂O yang
digunakan pada proses PCR



Pengaturan proses PCR pada
Thermalcycle



Produk PCR



Proses Elektroforesis



Visualisai gel agarose dengan UV
Transluminator



Produk PCR dikirim ke
PT.Genetica Science